BILLORA

Smart Shop Billing System

Project Report

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Python Project

A command-line billing system designed to organize shop, customer, product, billing, receipt, and bill-history operations.

1. Introduction

Billora is a Python-based smart shop billing system developed to make the basic billing process of a shop simple and organized. It allows shop details to be stored, customer and product information to be entered, discounts and billing amounts to be calculated, receipts to be generated, and previous bills to be viewed.

The project uses a modular Python structure, with separate files for different responsibilities, and JSON files for persistent shop and billing data.

2. Problem Statement

Small shops often need a simple way to manage customer details, product details, discounts, billing calculations, and previous bills without performing all calculations manually. Billora was developed as a command-line billing system to make this process more organized and reduce calculation errors during billing.

3. Objectives

• Organize shop and customer information.
• Record one or more products in a bill.
• Calculate discounts, final prices, total amount, and savings automatically.
• Generate a readable bill receipt.
• Store completed bills for later viewing.
• Validate important user inputs.

4. Functional Requirements

1. Shop management — store and view shop name, address, and license number.
2. Customer details — enter customer name and mobile number.
3. Product details — enter product name, quantity, MRP, and discount percentage.
4. Billing calculations — calculate discounted prices, grand total, and total savings.
5. Receipt generation — display a formatted receipt containing shop, customer, and product information.
6. Bill history — save completed bills and view previous records.
7. Input validation — validate quantity, MRP, discount percentage, customer name, and mobile number.

5. Non-Functional Requirements

• Usability: provides a simple command-line menu and readable receipt output.
• Reliability: stores billing information in JSON files for later retrieval.
• Maintainability: separates major responsibilities into individual Python modules.
• Error handling: validates important input values before continuing.

6. System Architecture

Billora follows a modular architecture. main.py acts as the main controller, while separate modules handle shop management, customer details, products, calculations, billing, receipt generation, validation, and storage.

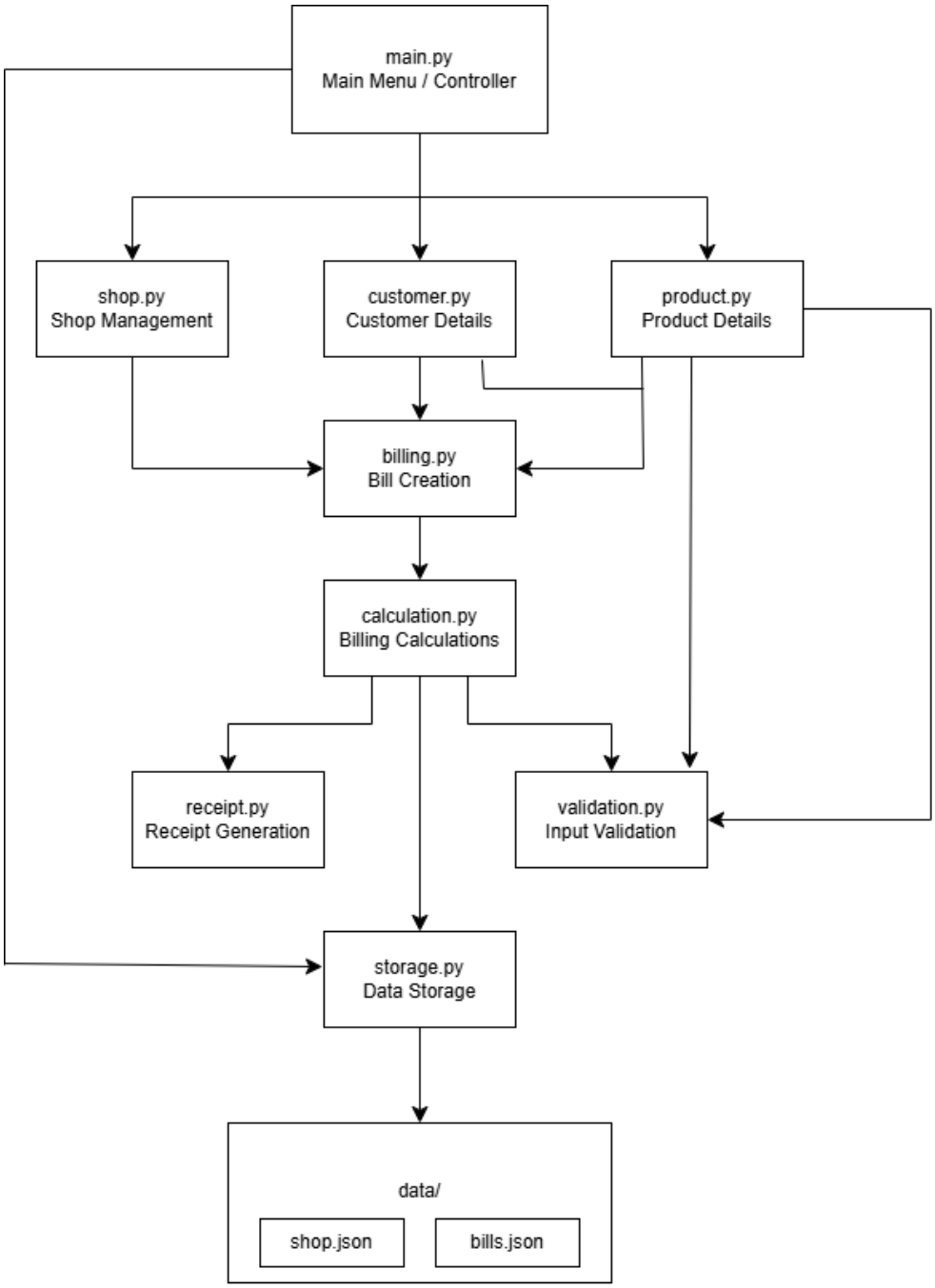


Figure 1: Billora system architecture

7. Billing Workflow

The workflow begins when Billora is opened and a new bill is selected. Customer and product information is entered and validated. The system then calculates the bill, stores the bill in bills.json, generates the receipt, and returns to the main menu.

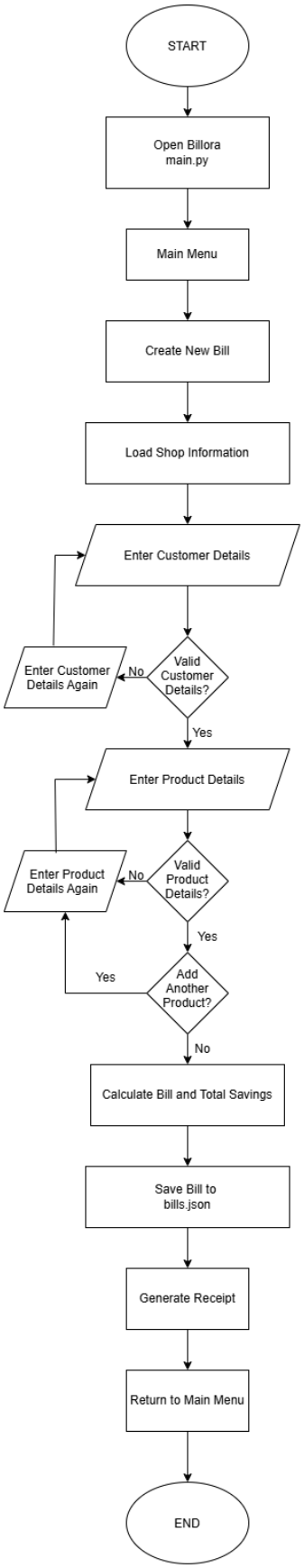


Figure 2: Billora billing workflow

8. Project Structure

main.py — Main menu and controller
shop.py — Shop management
customer.py — Customer details
product.py — Product details
calculation.py — Billing calculations
billing.py — Bill creation
receipt.py — Receipt generation
validation.py — Input validation
storage.py — JSON data storage
test_billora.py — Calculation tests
data/shop.json — Shop information
data/bills.json — Stored bills

9. Implementation and Results

The following screenshots demonstrate the working command-line application, including bill creation, receipt generation, bill history, shop management, and program exit.

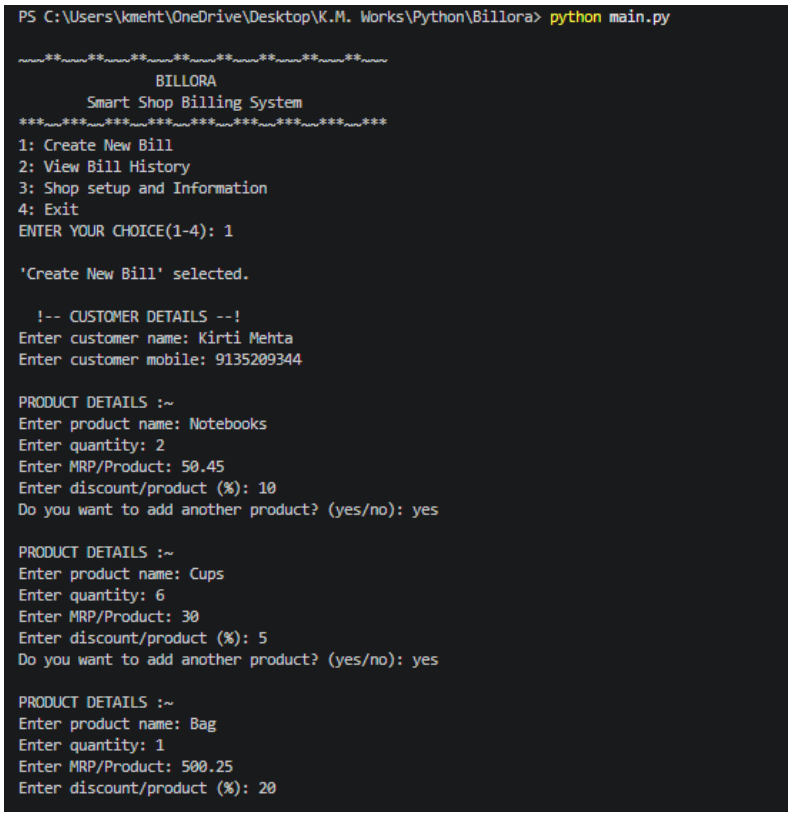


Figure 3: Bill creation and product entry

```
PRODUCT DETAILS :~
Enter product name: Bag
Enter quantity: 1
Enter MRP/Product: 500.25
Enter discount/product (%): 20
Do you want to add another product? (yes/no): no

^^^^^^**^^^^^^^^^^^^^^^^^^^^^^**^^^^^^^^^^^^^^
                BILLORA
            'SMART SHOP BILLING'
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Shop Name     : ABC Workshop
Shop Address  : Bhopal,MP,India
License Number: 23MBI110974

Customer Name : Kirti Mehta
Mobile Number : 9135209344

-------------------------------------------------------------------
Product        Quantity     MRP/Pc          Discount      Cost
-------------------------------------------------------------------
Notebooks      2 pcs        Rs. 50.45       10.0%         Rs. 90.81
Cups           6 pcs        Rs. 30.00       5.0%          Rs. 171.00
Bag            1 pcs        Rs. 500.25      20.0%         Rs. 400.20
-------------------------------------------------------------------
Grand Total : Rs. 662.01
Total Saved : Rs. 119.14

~~~**~~~**~~~**~~~**~~~**~~~**~~~**~~~**~~~
               BILLORA
       Smart Shop Billing System
***~~***~~***~~***~~***~~***~~***~~***~~***
1: Create New Bill
2: View Bill History
3: Shop setup and Information
4: Exit
ENTER YOUR CHOICE(1-4): 2
```

Figure 4: Generated Billora receipt

```
'Bill History' selected.

Customer Name : Mahesh
Mobile Number : 9876541230
Grand Total   : Rs. 180.00
Total Saved   : Rs. 20.00
----------------------------------------

Customer Name : Rakul preet
Mobile Number : 9865723140
Grand Total   : Rs. 105000.00
Total Saved   : Rs. 45000.00
----------------------------------------

Customer Name : Mina
Mobile Number : 9632015874
Grand Total   : Rs. 30.00
Total Saved   : Rs. 0.00
----------------------------------------

Customer Name : Kirti Mehta
Mobile Number : 9135209344
Grand Total   : Rs. 662.01
Total Saved   : Rs. 119.14
----------------------------------------

~~~**~~~**~~~**~~~**~~~**~~~**~~~**~~~**~~~
               BILLORA
       Smart Shop Billing System
***~~***~~***~~***~~***~~***~~***~~***~~***
1: Create New Bill
2: View Bill History
3: Shop setup and Information
4: Exit
ENTER YOUR CHOICE(1-4): 3

'Shop setup and Information' selected.
```

Figure 5: Bill history

```
      SHOP SETUP & INFORMATION
1: Set Up Your Shop
2: View Your Shop Information
3: Back to Main Menu
ENTER YOUR CHOICE(1-3): 1

|| SHOP SETUP ||
Enter your shop name: KM WorkHouse
Enter your shop address: Bhagalpur,Bihar,India
Enter your shop's license number: 26BAI10175

~~~**~~~**~~~**~~~**~~~**~~~**~~~**~~~
                BILLORA
        Smart Shop Billing System
***~~~***~~~***~~~***~~~***~~~***~~~***~~~***
1: Create New Bill
2: View Bill History
3: Shop setup and Information
4: Exit
ENTER YOUR CHOICE(1-4): 3

'Shop setup and Information' selected.

      SHOP SETUP & INFORMATION
1: Set Up Your Shop
2: View Your Shop Information
3: Back to Main Menu
ENTER YOUR CHOICE(1-3): 2

          SHOP INFORMATION
Shop Name     : KM WorkHouse
Shop Address  : Bhagalpur,Bihar,India
License Number: 26BAI10175
---x---x---x---x---x---x---x---x---x---x---
```

Figure 6: Shop setup and shop information

```
~~~**~~~**~~~**~~~**~~~**~~~**~~~**~~~**~~~
                BILLORA
        Smart Shop Billing System
***~~~***~~~***~~~***~~~***~~~***~~~***~~~***
1: Create New Bill
2: View Bill History
3: Shop setup and Information
4: Exit
ENTER YOUR CHOICE(1-4): 4

'Exit' selected.

THANK YOU for using BILLORA!
PS C:\Users\kmeht\OneDrive\Desktop\K.M. Works\Python\Billora>
```

Figure 7: Program exit

```
PS C:\Users\kmeht\OneDrive\Desktop\K.M. Works\Python\Billora> python test_billora.py
Discount calculation test passed.
Final price calculation test passed.
Total price calculation test passed.
Total savings calculation test passed.
```

Figure 8: Calculation tests

10. Testing

A separate test file, test_billora.py, was used to test the calculation functions. The following tests were successfully passed:

• Discount calculation test
• Final price calculation test
• Total price calculation test
• Total savings calculation test

The successful test output is included in the implementation screenshots.

11. Challenges and Learnings

During development, the project was divided into smaller modules so that each part had a clear responsibility. Working with JSON storage helped demonstrate persistent data handling. Input validation was added to prevent invalid quantities, MRP values, discount percentages, customer names, and mobile numbers. Testing the calculation functions also helped verify the billing logic independently.

12. Future Enhancements

Possible future improvements include a graphical user interface, product inventory management, search and filtering of bill history, additional reporting features, and a database-backed storage system.

13. Conclusion

Billora demonstrates a practical Python-based solution for a basic shop billing workflow. The project applies modular programming, functions, validation, calculations, JSON storage, receipt formatting, and testing to create a functional command-line application.

14. References

• Python documentation and standard library concepts used during development.
• VITyarthi project guidelines provided for the Build Your Own Project assignment.